

# Matias Freire

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## EDUCATION

### Stevens Institute of Technology

Hoboken, NJ

*Bachelor of Science in Computer Science, Minor in Math*

*Sept 2024 – May 2028*

**Relevant Coursework:** Algorithms, Data Structures, Discrete Structures, Computer Architecture and Organization, Multivariable Calculus, Vectors and Matrices, Probability, Mechanics, Biology and Biotechnology

**Awards:** STEP Alumni Scholarship: Selected as 1 of 3 recipients out of 200+ members for outstanding leadership

**Certifications:** Hugging Face AI Agents Fundamentals, Bloomberg Market Concepts

## EXPERIENCE

### Deep Learning Research Assistant/Intern

Aug 2025 – Present(ongoing)

*Advanced Quantum Materials Lab*

*Hoboken, NJ*

- Researched MoS defect characterization for deep learning studies and applied established preprocessing, synthetic data generation, and modeling techniques to analyze vacancy and polymorph defects within a unified research Dash-Plotly Hub.
- **Synthesized and presented nanomaterials research to Dr. Eui-Hyeok Yang** and others in the Lab through bi-weekly presentations, enhancing team-wide understanding of material properties and defect analysis methodologies.

### Machine Learning Ignite Fellow Financial Track

Aug. 2025 – January 2026

*AI4ALL*

*Remote*

- Completed an intensive applied machine learning fellowship, implementing key models such as linear regression and tree-based methods using **Pandas and scikit-learn**, while performing hands-on data preprocessing, feature engineering, and exploratory analysis on structured datasets
- Collaborated within a team to plan, develop, and present a final machine learning project, utilizing project management principles and ethical AI frameworks, with deployment and documentation executed in Jupyter Notebook and scikit-learn

## PROJECTS

### MoS2 Image Synthesis & Analysis Platform | *PyTorch, Dash-Plotly*

- Developed an end-to-end **Dash/Plotly platform for Deep learning Research** on STEM defects in MoS<sub>2</sub>, providing an intuitive, streamlined workflow for data generation and experimentation
- Automated synthetic STEM image generation with customizable defects, realistic noise injection, and **implemented a ResUNet** model for vacancy and polymorph classification

### Colorstack Stevens Chapter Website | *Next.js, React, Tailwind, Typescript*

- Built from scratch modular, responsive web pages using **Next.js, React, TypeScript, and Tailwind CSS** for the ColorStack Stevens chapter, **serving 60+ registered members** across the main website
- **Prototyped and validated** website architecture and UI components with fast HTML/CSS iterations in Cursor IDE, establishing a reusable design system before full implementation in Next.js and TypeScript

### Interactive-Neural-Network-Digit-Classifer | *PyTorch, Streamlit, Numpy, Pygame*

- Achieved 96% accuracy on MNIST digit recognition by building and training a 4-layer Convolutional Neural Network **in PyTorch**.
- Enabled real-time user interaction and visualization by integrating a PyGame drawing canvas and deploying the model with a Streamlit web interface

## LEADERSHIP ROLES & PROFESSIONAL DEVELOPMENT

### Head of Extern Relations & Co-Chair of Technology

Apr 2025 – Present(ongoing)

*Colorstack Stevens Chapter*

*Hoboken, NJ*

- Increased active membership by 50% and online engagement by 25% from Fall '24 to Fall '25 by spearheading social media strategy and organizing 14+ events per semester.
- Strengthened member coding proficiency by collaborating with peers to solve 32+ data structure problems each term

### STEP Tutor-Counselor/Resident Assistant (TCRA)

Jun 2025 – Aug 2025

*STEP Bridge Summer Program*

*Hoboken, NJ*

- Strengthened STEM readiness for 50+ incoming freshmen by leading evening Calculus I/II and Physics recitations with targeted problem-solving support to build technical proficiency and conceptual understanding in fast-paced, credit-bearing courses.
- Provided personalized Calculus tutoring to +25 students weekly, including late-night academic support, while supporting residential life operations as a **Resident Assistant**

## TECHNICAL SKILLS

**Programming Languages:** Java, Python, C/C++, JavaScript

**Web Development:** TypeScript, React, Tailwind, Next.js

**Developer Tools:** Git, GitHub, VS Code, Jupyter Notebooks

**Libraries:** Dash, Pandas, NumPy, PyTorch, Scikit-learn, Streamlit